

Course Code : ECT 451-2

KQLR/RS – 24 / 1673

Seventh Semester B. Tech. (Electronics and Communication Engineering) Examination

MICROWAVE THEORY AND TECHNIQUES

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
 - (2) All questions carry marks as indicated.
 - (3) Illustrate your answer with neat sketches.
 - (4) Assume suitable data wherever necessary.
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1. (a) Justify, why Microwave frequency bands are preferred over low frequency band. 5(CO1)
 - (b) Enlist various Microwave frequency band applications in different fields according to its property. 5(CO1)
 2. (a) Discuss single stub Impedance matching techniques used in transmission lines along with merits and demerits. 5(CO2)
 - (b) Estimate the band over which WR90 waveguide can be used near to dominant mode. 5(CO2)
 3. (a) Summarize the working principle of Rat - Race Junction along with its applications. 5(CO4)
 - (b) Prove that H – Plane tee work as a 3 dB power splitter. 5(CO4)
 4. (a) Interpret working principle of Electronic Mixer. 5(CO3)
 - (b) Design a Rectangular Microstrip Antenna (RMSA) at 2.3 GHz. Choose a substrate with dielectric constant of 2.2, loss tangent of 0.02 and thickness of 1.6 mm. 5(CO3)

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5. (a) Illustrate Resonant Cavity frequency measurement Technique in detail. 5(CO5)
- (b) Return loss for an antenna is found out to be -13 dBi on VNA at one resonant frequency. Estimate VSWR, Reflection Coefficient due to the Antenna. 5(CO5)
6. (a) Draw and discuss the basic block diagram of RADAR system. 5(CO1)
- (b) Determine the cross section area of the target that can be sighted by a marine RADAR operating at 20 GHz; it has a maximum range of 50 KM with an antenna gain of 5000 . If the transmitter has a power of 350 KW and minimum detectable signal of 10^{-12} W. 5(CO1)

